ModelIt!

Teacher's Guide

What Makes Things Move?

Pushes and Pulls All Around Us

Quick Reference

Grade Level	Kindergarten
NGSS Standard	K-PS2-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: K-PS2-2

Analyze data to determine if a design solution works as intended to change the speed or direction of an object with a push or a pull.

Three-Dimensional Learning

Science Practice	**Analyzing and Interpreting Data** — Students compare how far the car goes with a gentle push versus a big push, analyzing simple data about distance.
Disciplinary Core Idea	**PS2.A Forces and Motion** — Pushes and pulls can have different strengths and directions. A bigger push or pull makes things speed up more.
Crosscutting Concept	**Cause and Effect** — Students observe that changing push strength (cause) changes how far the car travels (effect).

CAST Testing Alignment

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Forces are everywhere in our daily lives. Every time something starts moving, stops, or changes direction, a force is responsible. Understanding pushes and pulls helps young learners make sense of how the world works.

What Are Pushes and Pulls?

A push is a force that moves an object away from the source of the force. A pull is a force that moves an object toward the source. These are the two basic types of forces. Opening a door can be a push or a pull. Kicking a ball is a push. Dragging a backpack is a pull. Forces can be big or small, and the size of the force affects how much the object moves.

How Force Affects Motion

The strength of a push or pull determines how much an object's motion changes. A gentle push on a toy car makes it roll slowly and stop nearby. A strong push makes it zoom across the room. This is related to Newton's Second Law of Motion — the bigger the force, the greater the acceleration. At the kindergarten level, students simply observe that 'bigger pushes make things go farther and faster.'

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System
Students identify Push Strength as the external component (we choose how hard to push) and Distance Traveled as the internal component (it changes based on how hard we pushed).

E — Establish Relationships
Students discover that stronger Push Strength leads to greater Distance Traveled — a positive relationship.

V — Visualize & Model
Students build a simple two-part model connecting Push Strength to Distance Traveled with a positive arrow.

E — Evaluate Outcomes
Students test Gentle Push vs. Big Push and observe the distance difference.

R — Revise & Extend
Students explore what happens on different surfaces (carpet vs. tile) to see how the floor changes things.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"What makes things move? Can something start moving all by itself? Let us find out!"*

Watch For

Place a toy car on a table. Wait. Ask: Why is it not moving? Then push it. Ask: What did I do?

Slide 2: What Will We Learn?

3 min

Say: *"Today we will learn about pushes and pulls! A push moves things away from you. A pull brings things toward you. Together, they are called FORCES."*

Watch For

Practice push gesture (hands pushing away) and pull gesture (hands pulling toward chest). Say the words together.

Slide 3: Let's Be Scientists!

3 min

Say: *"Our big question is: What makes a toy car start moving? Can it move by itself? What does it need?"*

Watch For

Students turn to a partner and share one thing they push and one thing they pull every day.

Slide 4: What Are the Parts?

5 min

Say: *"Our model has two parts. Push Strength is how hard we push — WE choose that! Distance Traveled is how far the car goes — that happens on its own after we push."*

Watch For

Use sorting mats. Students place Push Strength OUTSIDE and Distance Traveled INSIDE. Practice gentle push and big push motions.

Slide 5: How Are They Connected?

5 min

Say: *"When Push Strength goes UP, what happens to Distance Traveled? The car goes FARTHER! That is a POSITIVE connection."*

Watch For

Demonstrate: push car gently (short distance), push car hard (long distance). Students give a thumbs up for positive.

Slide 6: What Did We Find Out?

5 min

Say: *"We discovered that bigger pushes make things go farther! Pushes and pulls are called forces, and forces make everything move."*

Watch For

Chant together: 'When Push Strength goes up, Distance goes up!' Complete the sentence frame as a class.

Slide 7: STEM Challenge

10 min

Say: *"Now you are the engineers! Can you push your car so it stops RIGHT on the finish line? Not too far, not too short — just right!"*

Watch For

Place tape finish lines at different distances. Students practice pushing with different strengths. Record results with stickers (too far, too short, just right).

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Push Strength
- **INTERNAL:** Distance Traveled

Relationships

- **Push Strength to Distance Traveled:** POSITIVE (+) — When you push harder, the car goes farther because a stronger force gives the car more energy to keep rolling.

Key Discoveries

- A bigger push makes the car go farther across the floor
- A tiny push makes the car go only a little bit
- Pushes and pulls are forces — they make things start moving, stop, or change direction

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Things move by themselves"	Nothing moves without a force acting on it. A ball rolling across the floor was pushed by someone or something. Wind, gravity, and people all provide forces.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Heavier things are harder to push so they never go as far"	While heavier objects need more force to start moving, once in motion they can actually travel far because their weight helps them keep going. The key factor is the force applied, not just the weight.